

Mathematics A

Unit 1 • Chapter OL-T48 Classified

Cross-session • chapter-organised candidate

21 classified items • 75 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Unit 1 • Chapter OL-T48 • 21 items • 75 marks

Chapter	Items	Marks	Starts
01. OL-T48 Right-Angled Triangles - Pythagoras & Trigonome	21	75	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Right-Angled Triangles - Pythagoras & Trigonometry

Unit 1 • 75 marks • 21 item(s)

June 2025 | 4MA1-2H Q10 | 5 marks

Use two connected right triangles

June 2024 | 4MA1-1H Q11 | 4 marks

Use a right-triangle result in a practical threshold or cost

June 2024 | 4MA1-2H Q5 | 3 marks

Find opposite side with sine or tangent

November 2023 | 4MA1-2H Q2 | 4 marks

Use two connected right triangles

June 2023 | 4MA1-1H Q5 | 4 marks

Combine right-triangle trig with area or perimeter

June 2023 | 4MA1-2H Q12 | 4 marks

Use two connected right triangles

November 2024 | 4MA1-1H Q12 | 4 marks

Use two connected right triangles

November 2024 | 4MA1-2H Q3 | 3 marks

Find adjacent side with cosine or tangent

June 2022 | 4MA1-2H Q6 | 3 marks

Find adjacent side with cosine or tangent

January 2022 | 4MA1-1H Q3 | 3 marks

Find opposite side with sine or tangent

January 2022 | 4MA1-2H Q14 | 3 marks

Use a right-triangle result in a practical threshold or cost

January 2020 | 4MA1-1H Q9 | 3 marks

Find opposite side with sine or tangent

January 2021 | 4MA1-1H Q10 | 4 marks

Combine right-triangle trig with area or perimeter

June 2021 | 4MA1-2H Q7 | 3 marks

Find hypotenuse with sine or cosine

November 2021 | 4MA1-1H Q2 | 3 marks

Find a shorter side by Pythagoras

... 6 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 17

4MA1-2H Q10

UNIT 1
5 marks

- 10 ABD and ABC are right-angled triangles.

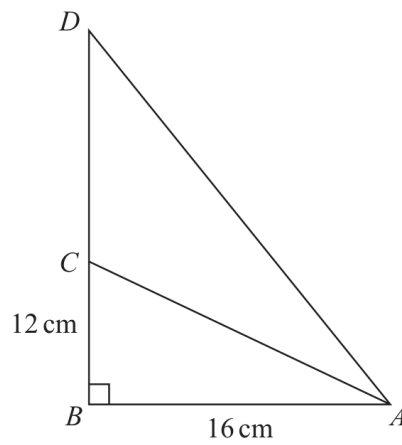


Diagram **NOT**
accurately drawn

$$AB = 16 \text{ cm} \quad BC = 12 \text{ cm} \quad AD = 1.5 \times AC$$

Find the length of CD

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 10 is 5 marks)

WORKING SPACE

Question 07

4MA1-1H Q11

MODULAR UNIT 1

4 marks

- 11 The diagram shows a roof support.

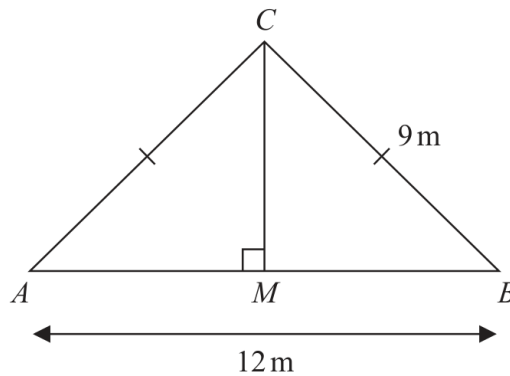


Diagram **NOT**
accurately drawn

The roof support is made from four lengths of wood, AB , AC , BC and MC

$$AC = BC = 9 \text{ m} \quad AB = 12 \text{ m}$$

$$\text{angle } AMC = 90^\circ$$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.

Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.

Show your working clearly.

..... euros

Working space for Question 07

Continue your method

UNIT 1

4 marks

WORKING SPACE

Question 19

4MA1-2H Q5

MODULAR UNIT 1

3 marks

- 5 The diagram shows triangle ABC

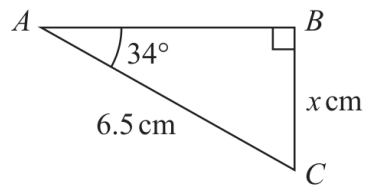


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

WORKING SPACE

Question 15

4MA1-2H Q2 whole

MODULAR UNIT 1

4 marks

- 2 The diagram shows triangle ABC and triangle ECD

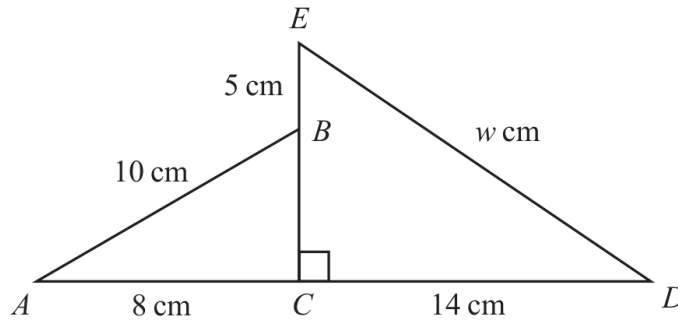


Diagram **NOT**
accurately drawn

ACD and EBC are straight lines.

$$AB = 10 \text{ cm} \quad AC = 8 \text{ cm} \quad EB = 5 \text{ cm} \quad CD = 14 \text{ cm} \quad ED = w \text{ cm}$$

Work out the value of w

Give your answer correct to one decimal place.

$w = \dots\dots\dots$

WORKING SPACE

Question 01

4MA1-1H Q5

MODULAR UNIT 1

4 marks

- 5 The diagram shows a shaded shape $AEBCD$ made by removing triangle AEB from rectangle $ABCD$

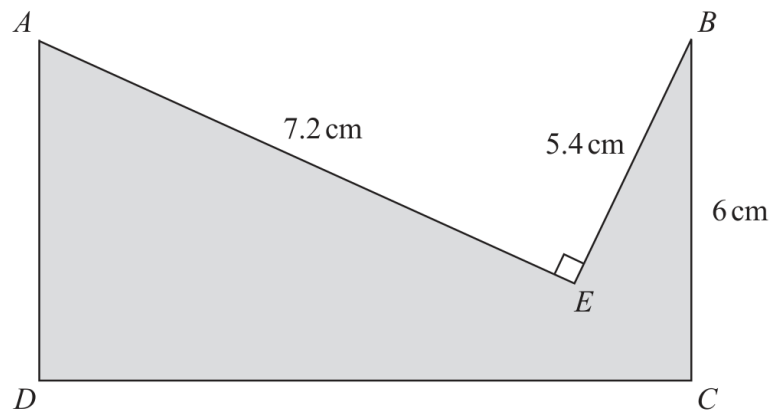


Diagram **NOT**
accurately drawn

$$AE = 7.2 \text{ cm} \quad BE = 5.4 \text{ cm} \quad BC = 6 \text{ cm} \quad \text{angle } AEB = 90^\circ$$

Work out the perimeter of the shaded shape.

..... cm

WORKING SPACE

Question 17

4MA1-2H Q12

MODULAR UNIT 1

4 marks

12 The diagram shows right-angled triangle ABD

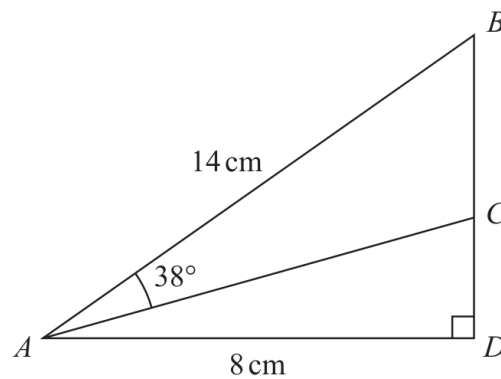


Diagram **NOT**
accurately drawn

$$AB = 14 \text{ cm} \quad AD = 8 \text{ cm}$$

C is the point on BD such that angle $BAC = 38^\circ$

Work out the length of CD

Give your answer correct to 3 significant figures.

..... cm

WORKING SPACE

Question 09

4MA1-1H Q12 M01

MODULAR UNIT 1

4 marks

- 12 ABC is a right-angled triangle.
 D is a point on BC

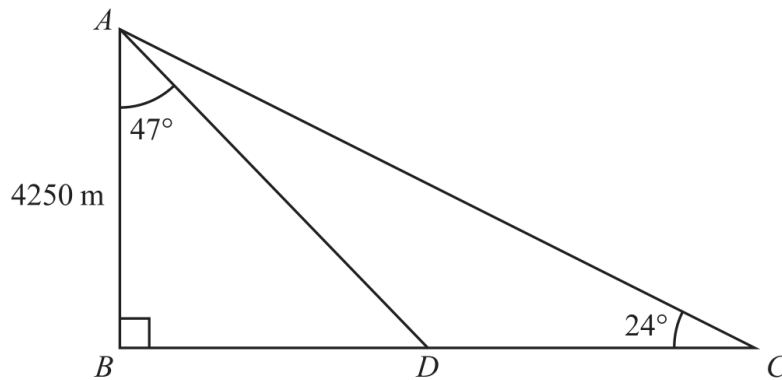


Diagram **NOT**
accurately drawn

$$AB = 4250 \text{ m} \quad \text{angle } BAD = 47^\circ \quad \text{angle } BCA = 24^\circ$$

Work out the length of DC

Give your answer correct to the nearest integer.

..... m

WORKING SPACE

Question 20

4MA1-2H Q3 M01

MODULAR UNIT 1

3 marks

- 3 The diagram shows triangle PQR

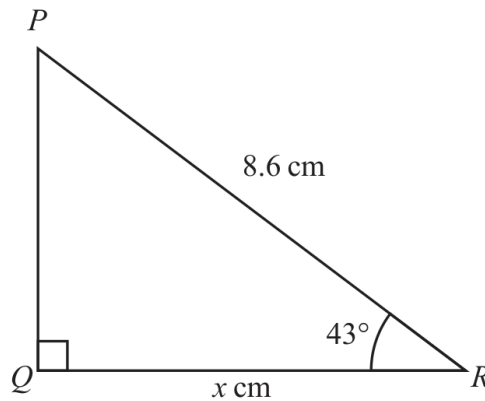


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

WORKING SPACE

Question 19

4MA1-2H Q6 Q6

MODULAR UNIT 1

3 marks

- 6 The diagram shows triangle PQR .

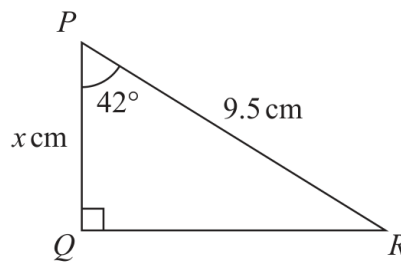


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

WORKING SPACE

Question 04

4MA1-1H Q3

MODULAR UNIT 1

3 marks

- 3 The diagram shows triangle ABC

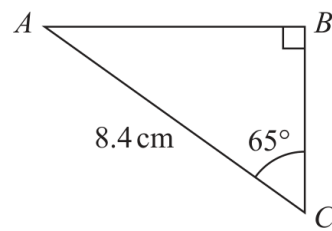


Diagram **NOT**
accurately drawn

Work out the length of the side AB
Give your answer correct to 3 significant figures.

..... cm

WORKING SPACE

Question 24

4MA1-2H Q14

MODULAR UNIT 1

3 marks

- 14 A zip wire is shown as the dashed line AC in the diagram.

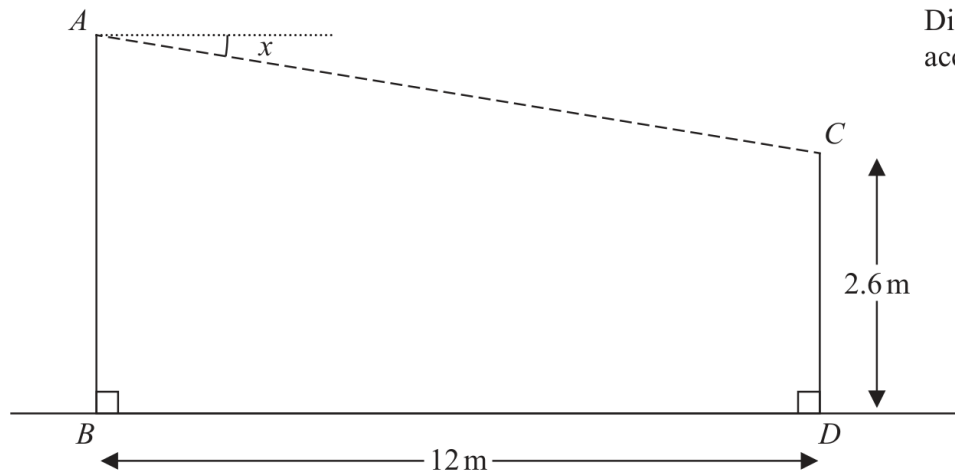


Diagram **NOT**
accurately drawn

The zip wire is supported by two vertical posts AB and CD standing on horizontal ground.

$$CD = 2.6 \text{ m} \quad BD = 12 \text{ m}$$

The zip wire makes an angle x with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle x to be at least 5°

Work out the least possible height of the post AB
Give your answer correct to 3 significant figures.

..... m

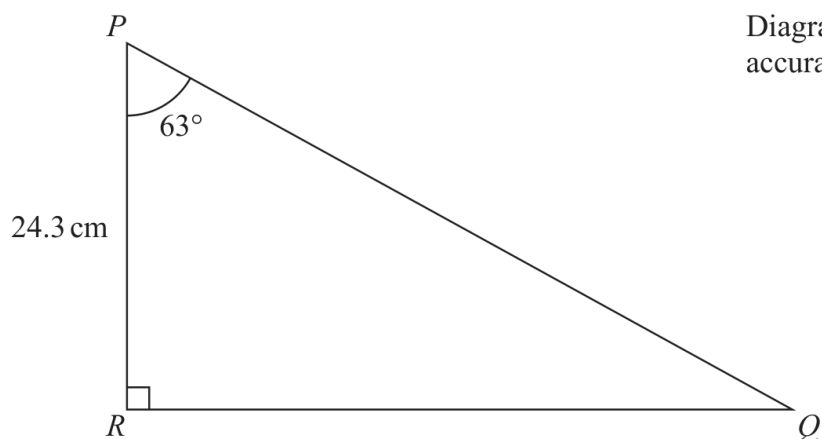
Working space for Question 24

Continue your method

UNIT 1
3 marks

WORKING SPACE

9 Here is a right-angled triangle.



Calculate the length of PQ .
Give your answer correct to 3 significant figures.

..... cm

YOUR WORKING

10 Here is isosceles triangle ABC .

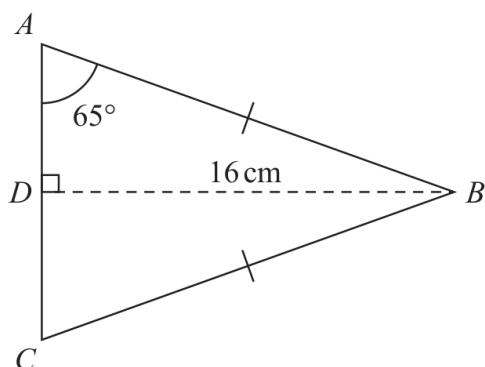


Diagram **NOT**
accurately drawn

D is the midpoint of AC and $DB = 16$ cm.

Angle $DAB = 65^\circ$

Work out the perimeter of triangle ABC .

Give your answer correct to one decimal place.

..... cm

YOUR WORKING

7 Here is a right-angled triangle.

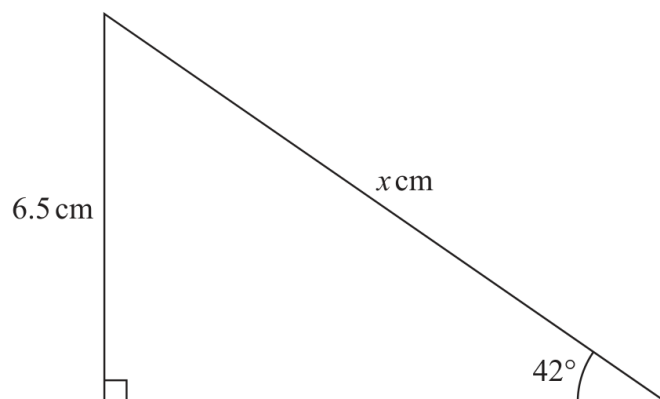


Diagram **NOT**
accurately drawn

Work out the value of x .
Give your answer correct to one decimal place.

$x = \dots\dots\dots$

YOUR WORKING

2 Here is a right-angled triangle.

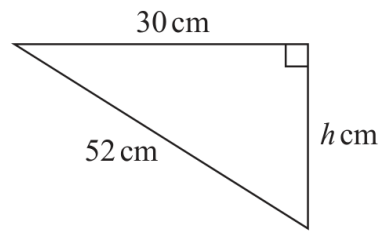


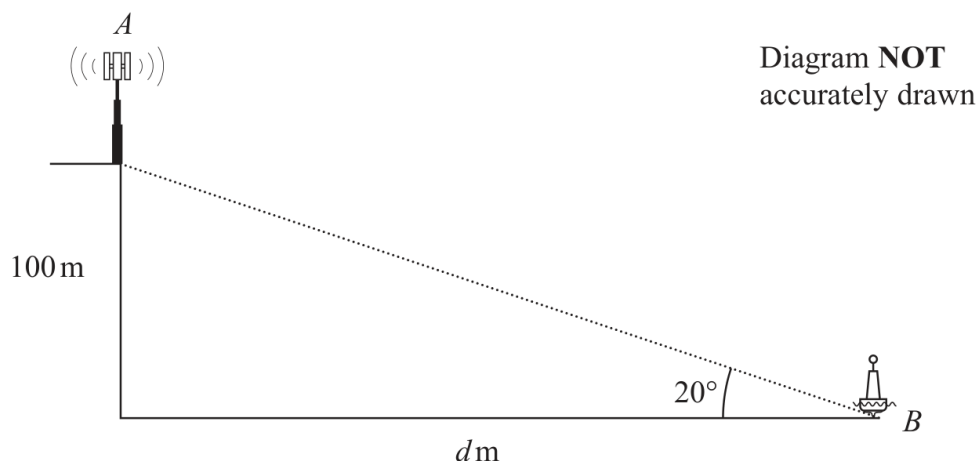
Diagram **NOT**
accurately drawn

Calculate the value of h .
Give your answer correct to 3 significant figures.

$h =$

YOUR WORKING

- 12 The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.



The height of the cliff is 100 metres.

The buoy is at the point B that is d metres from the base of the cliff.

The angle of elevation from B to the top of the cliff is 20°

- (a) Calculate the value of d .

Give your answer correct to 3 significant figures.

$d = \dots\dots\dots$
(3)

The point A at the top of the radio mast is vertically above the top of the cliff.

The angle of elevation from B to A is 25°

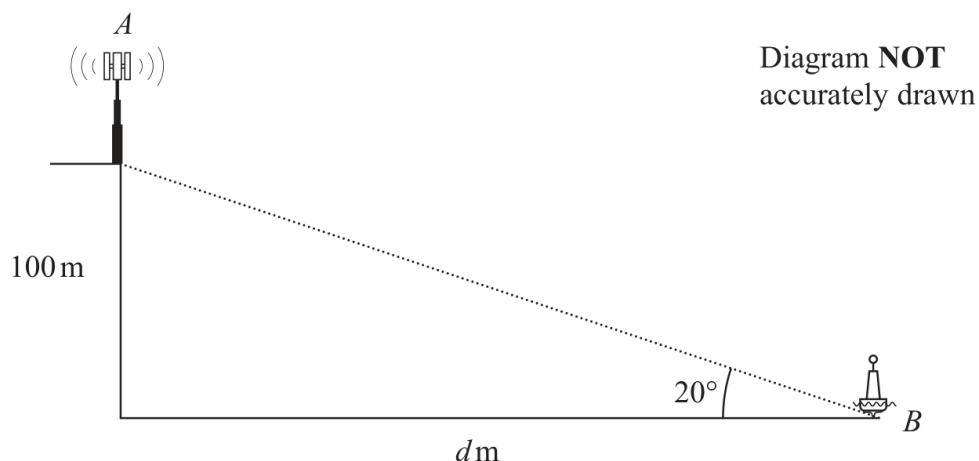
- (b) Calculate the height of the radio mast.

Give your answer correct to 3 significant figures.

$\dots\dots\dots$ m
(3)

YOUR WORKING

- 12 The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.



The height of the cliff is 100 metres.

The buoy is at the point B that is d metres from the base of the cliff.

The angle of elevation from B to the top of the cliff is 20°

- (a) Calculate the value of d .

Give your answer correct to 3 significant figures.

$d = \dots\dots\dots$
(3)

The point A at the top of the radio mast is vertically above the top of the cliff.

The angle of elevation from B to A is 25°

- (b) Calculate the height of the radio mast.

Give your answer correct to 3 significant figures.

$\dots\dots\dots$ m
(3)

YOUR WORKING

- 7 The diagram shows a right-angled triangle.

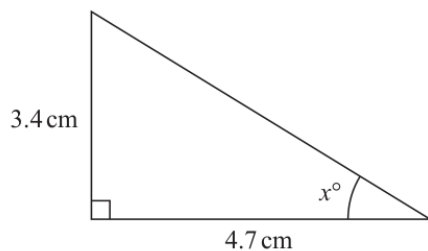


Diagram **NOT**
accurately drawn

Calculate the value of x .
Give your answer correct to one decimal place.

$x =$

(Total for Question 7 is 3 marks)

YOUR WORKING

- 10 The diagram shows an isosceles triangle, with base length 24 cm.

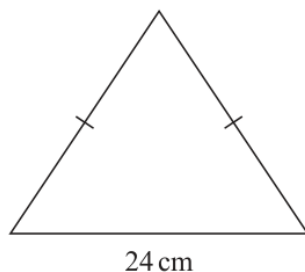


Diagram **NOT**
accurately drawn

The perimeter of the triangle is 54 cm.

Work out the area of the triangle.

..... cm²

(Total for Question 10 is 5 marks)

YOUR WORKING

11 ABC is a right-angled triangle.

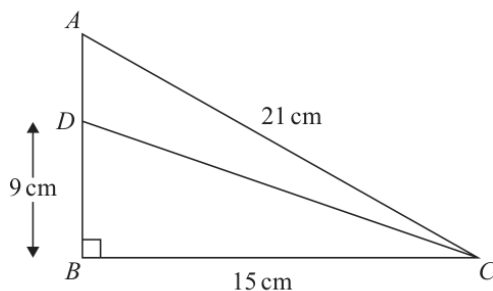


Diagram **NOT**
accurately drawn

$$AC = 21 \text{ cm} \quad BC = 15 \text{ cm} \quad \text{angle } ABC = 90^\circ$$

The point D lies on AB such that $DB = 9 \text{ cm}$

Work out the size of angle ACD

Give your answer correct to one decimal place.

(Total for Question 11 is 4 marks)

YOUR WORKING

11 The diagram shows a right-angled triangle ABC

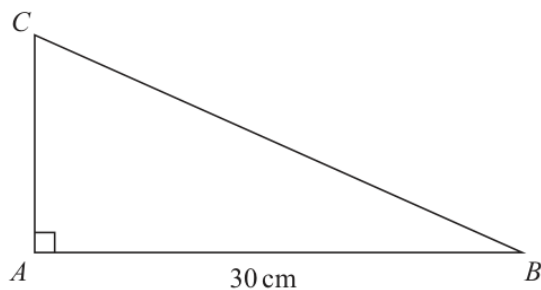


Diagram **NOT**
accurately drawn

Area of triangle $ABC = 240 \text{ cm}^2$

Work out the perimeter of triangle ABC

..... cm

(Total for Question 11 is 4 marks)

YOUR WORKING
